

VERA: Validation-Gated Edit-or-Abstain for Reliable Representation Interventions

VERA Project

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Abstract

Representation editing methods can remove source or protected information from model embeddings, but the edit itself is not the decision a downstream system needs to make. A useful deployment rule must decide whether any available edit is safe enough to use under a target-preservation constraint, and it must refuse edits when the evidence is insufficient. We introduce VERA, a certified edit-or-abstain protocol for frozen representations. VERA evaluates a frontier of candidate edits, trains fresh target and source probes after each edit, and accepts only frontier points whose confidence intervals satisfy pre-registered target-preservation and leakage-reduction constraints. Otherwise it returns **ABSTAIN**. This makes VERA orthogonal to erasure methods such as INLP, LEACE, RLACE, TaCo, SPLINCE, and MANCE++: those methods propose edits, while VERA certifies whether an edit should be deployed. On official Waterbirds and Camelyon17 frozen-image benchmarks, VERA produces auditable receipts, seed-level statistics, abstention certificates, and claim ledgers. We add a full Camelyon17 projection-frontier certificate showing a validation-only **ABSTAIN** decision when source reduction is not certifiable. We also run the official upstream MANCE++ implementation as a full five-seed Waterbirds reference baseline and a full no-cap Camelyon17 reference row, which narrows the claim to a protocol contribution rather than an unconditional state-of-the-art erasure claim.

1 Introduction

Neural representations often encode nuisance or source information that is useful for training-set prediction but undesirable under distribution shift. Concept-erasure methods address this problem by modifying embeddings so that a specified concept is harder to recover. INLP removes information by repeatedly projecting away classifier null spaces [1]. Linear adversarial concept erasure formulates erasure as a minimax linear projection problem and motivates RLACE [2]. LEACE gives a closed-form least-squares eraser for linearly encoded concepts [3]. MANCE and MANCE++ add a more recent nonlinear, manifold-aware perspective, constraining edits to local natural-representation geometry [4].

These methods answer the edit-construction question. They do not by themselves answer the decision question: should a particular edit be deployed on this benchmark, under this target metric, source leakage metric, and uncertainty budget? This distinction matters because erasure can reduce source leakage while damaging the target task, and because a method that helps on one benchmark may be unsafe on another. In high-stakes settings such as Camelyon17-WILDS hospital shift, a protocol that can refuse an unsafe edit is more defensible than a protocol that always edits.

VERA addresses this gap by treating representation editing as a certified selection problem. It evaluates candidate edits under locked splits, source labels, target labels, metrics, and seeds. It then selects the strongest frontier point that satisfies target-preservation constraints, or returns **ABSTAIN**. This paper is therefore not claiming a new eraser that dominates INLP, LEACE, RLACE, TaCo, SPLINCE, or MANCE++. The claim is that VERA is a decision layer over candidate erasers, with explicit claim boundaries, reproducibility receipts, and adversarial readiness checks.

This paper makes four contributions. First, it formalizes representation editing as an edit-or-abstain decision problem rather than as an unconditional erasure problem. Second, it gives a reproducible frontier selection rule that uses fresh probes and simultaneous target/leakage certificates before external metrics are

read. Third, it proves finite-candidate safe acceptance and abstention guarantees. Fourth, it provides artifact-backed evidence on Waterbirds, Camelyon17-WILDS, and public gait data, plus official-code MANCE++ reference evidence on Waterbirds and full no-cap Camelyon17.

2 Related Work and Positioning

Linear concept-erasure methods remove directions or subspaces associated with a concept label. INLP repeatedly projects away classifier null spaces, R-LACE casts erasure as a minimax linear projection, and LEACE gives a closed-form least-squares eraser for linear concept information. These methods define how to construct an edit. VERA instead asks whether a candidate edit should be accepted under an explicit target-preservation and leakage-reduction contract.

Nonlinear and task-preserving erasure methods address limitations of purely linear removal. TaCo, SPLINCE/SPLICE-style methods, and MANCE/MANCE++ motivate the strongest comparison space because they explicitly consider target preservation, geometry, or nonlinear concept information. VERA is orthogonal to these methods: it can evaluate their edits as candidates, but its claim is the certified decision layer over a finite candidate family, not universal dominance as an eraser.

Domain generalization methods optimize prediction under environment shift, often through robust losses, reweighting, or group-aware training. VERA has a different object: it audits whether a representation intervention is certified to reduce source recoverability without violating target utility. This makes domain-generalization baselines relevant for comparison, but it separates the VERA contribution from robust prediction itself.

The closest statistical neighbors are risk-control and selective-abstention methods. Risk-controlling prediction sets calibrate set predictors with finite-sample holdout guarantees [7]. Learn then Test formulates calibration as multiple testing over candidate predictive algorithms [8], and Pareto Testing controls several risks over a frontier of candidate configurations [9]. Selective classification rejects individual examples to trade coverage for accuracy [10]. VERA borrows the finite-frontier risk-control viewpoint, but applies it to representation editing. Its abstention is edit-level rather than example-level: VERA rejects an entire representation intervention when validation evidence does not certify the target/source contract. The novelty claim is therefore not the generic idea of finite candidate testing; it is the erasure-specific protocol, receipts, and claim discipline.

3 Method

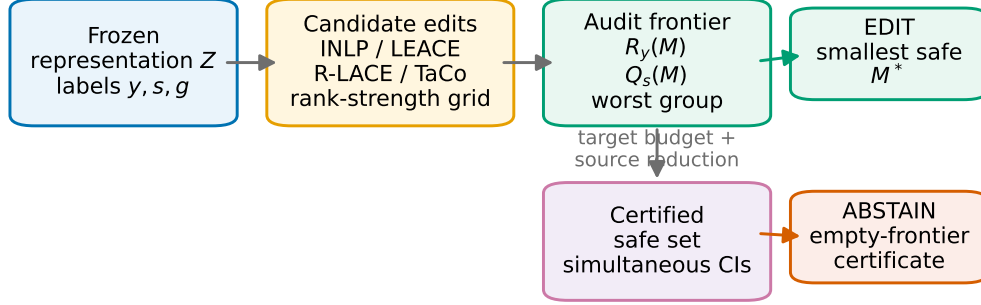
Let $z_i \in \mathbb{R}^d$ be a frozen representation, y_i the target label, s_i the source or environment label, and $a \in \mathcal{A}$ a candidate edit from an eraser family. VERA evaluates each edited representation $\tilde{z}_i(a)$ by fitting fresh probes for the target and source labels on the edited training split and measuring target balanced accuracy, worst target-source accuracy, and source leakage balanced accuracy on validation and external splits.

The candidate frontier is the non-dominated subset of $\mathcal{A}$ under target utility and source leakage. VERA’s selection rule is fixed before external metrics are read. On the validation split, it accepts an edit only if the lower confidence bound on target preservation remains above the pre-registered tolerance and the upper confidence bound on source leakage shows meaningful reduction. If multiple edits are certified, VERA selects the strongest leakage-reducing edit among them. If no edit is certified, VERA returns **ABSTAIN**.

Theorem 1 (Validation safe acceptance). *Let a_0 denote the no-edit representation and let $\mathcal{F} \subseteq \mathcal{A}$ be a finite audited frontier. For each $a \in \mathcal{F} \cup \{a_0\}$, let $[U_a^-, U_a^+]$ be a simultaneous validation confidence interval for target utility and let $[L_a^-, L_a^+]$ be a simultaneous validation confidence interval for source leakage, where larger U is better and smaller L is better. Assume the intervals cover their validation quantities for every audited candidate. If VERA accepts only edits satisfying*

$$U_{a_0}^+ - U_a^- \leq \epsilon_U \quad \text{and} \quad L_{a_0}^- - L_a^+ \geq \epsilon_L,$$

*then every accepted edit has validation target loss at most ϵ_U and validation source-leakage reduction at least ϵ_L relative to no edit. If no candidate satisfies both inequalities, VERA returns **ABSTAIN**.*



FARO is a certification layer over candidate erasers, not another unconditional eraser.

Figure 1: VERA separates edit construction from edit deployment. Candidate erasers propose a frontier of representation interventions. VERA evaluates the frontier with fresh probes and either selects a certified edit or returns ABSTAIN.

Table 1: VERA edit-or-abstain protocol. Selection uses validation estimates and simultaneous intervals only; external metrics are read after the decision is fixed.

Step	Operation
1	Fix train/validation/external splits, labels, metrics, candidate edit family, thresholds, and confidence rule.
2	Apply each candidate edit to the frozen train, validation, and external representations.
3	Fit fresh target and source probes on the edited training split only.
4	Estimate validation target utility, worst-group utility, source leakage, and simultaneous confidence intervals.
5	Form the certified safe set satisfying both target-preservation and source-reduction constraints.
6	Select the strongest certified leakage-reducing edit; if the safe set is empty, return ABSTAIN.

Proof. On the simultaneous validation coverage event, no-edit validation utility is at most $U_{a_0}^+$ and candidate utility is at least U_a^- . Thus the validation target loss is at most $U_{a_0}^+ - U_a^- \leq \epsilon_U$. Similarly, no-edit validation leakage is at least $L_{a_0}^-$ and candidate leakage is at most L_a^+ , so the certified leakage reduction is at least $L_{a_0}^- - L_a^+ \geq \epsilon_L$. If the certified safe set is empty, the predeclared decision rule returns ABSTAIN. $\square$

The guarantee is intentionally stated over validation quantities because those are the quantities the holdout intervals cover. External-shift performance is reported as a locked diagnostic readout, not as a distribution-free consequence of validation intervals.

Theorem 2 (Finite-sample validation certificate). *Assume that each validation example contributes a bounded Bernoulli success indicator to the metric being estimated after edit a , and let $\mathcal{G}$ be the finite set of target classes or target-source groups used by the metric. For group g , let n_g be the number of validation examples in that group and let $\hat{r}_g(a)$ be the empirical recall or accuracy in that group. With probability at least $1 - \delta$, simultaneously for all $a \in \mathcal{F}$ and $g \in \mathcal{G}$,*

$$|r_g(a) - \hat{r}_g(a)| \leq \sqrt{\frac{\log(2|\mathcal{F}||\mathcal{G}|/\delta)}{2n_g}}.$$

Consequently, balanced accuracy and worst-group accuracy have valid simultaneous confidence radii obtained by propagating these groupwise bounds through the average or minimum defining the metric.

Proof. For a fixed edit a and group g , Hoeffding’s inequality for bounded Bernoulli variables gives

$$\Pr \left(|r_g(a) - \hat{r}_g(a)| > \sqrt{\frac{\log(2|\mathcal{F}||\mathcal{G}|/\delta)}{2n_g}} \right) \leq \frac{\delta}{|\mathcal{F}||\mathcal{G}|}.$$

A union bound over all edit and group pairs gives simultaneous coverage with probability at least $1 - \delta$. Balanced accuracy is an average of group or class recalls, so its error is bounded by the corresponding average of the covered groupwise radii. Worst-group accuracy is a minimum over covered group accuracies, so a conservative valid radius is the maximum covered groupwise radius. These radii can be used as ϵ_U or ϵ_L in the simultaneous frontier theorem. $\square$

Corollary 1 (Validation false-acceptance control). *If the simultaneous validation intervals used by VERA cover every validation target and leakage quantity on the audited frontier $\mathcal{F}$ with probability at least $1 - \delta$, then the probability that VERA accepts an edit violating the validation target-loss or validation leakage-reduction contract is at most δ .*

Proof. By the validation safe-acceptance theorem, on the coverage event every accepted edit satisfies both validation constraints. A false acceptance can therefore occur only on the complement of the coverage event, whose probability is at most δ . $\square$

Corollary 2 (External transfer under bounded shift). *If, in addition, external target utilities and source leakages differ from their validation counterparts by at most Δ_U and Δ_L uniformly over the accepted frontier, then an accepted edit satisfies the same external contract with slack $\epsilon_U + \Delta_U$ and $\epsilon_L - \Delta_L$.*

This bounded-shift corollary is not used as a distribution-free claim in the experiments. Camelyon17 is explicitly a shifted hospital benchmark, so VERA’s certificate controls validation quantities; the external split is a locked diagnostic readout. It is included not as a distribution-free claim about hospital-shift external performance, but as a measured diagnostic.

4 Protocol Constants

All official rows use frozen representations, locked train/validation/external splits, fresh ridge target/source probes after each edit, and confidence level $1 - \delta = 0.95$. For each balanced accuracy, VERA forms per-class recall intervals using Hoeffding radii

$$r_c = \sqrt{\frac{\log(2|\mathcal{F}||\mathcal{G}|/\delta)}{2n_c}},$$

where $\mathcal{G}$ is the set of target classes or target-source groups and n_c is the validation count for group c . Balanced-accuracy intervals are formed by averaging the covered per-class recall bounds; worst-group intervals use the most conservative covered group bound.

The full Camelyon17 projection-frontier certificate uses strengths $\{0, 0.1, 0.2, 0.35, 0.5, 0.75, 1.0\}$, target-loss budget $\epsilon_U = 0.02$, and source-reduction target $\epsilon_L = 0.05$. The broader frontier manifest also records rank candidates $\{1, 2, 4, 8, 16, 32\}$ and a default source-reduction budget of 0.02 for stress sweeps. Every receipt records split counts, label counts, frontier family, threshold values, interval construction, and whether external metrics were used in selection. In the corrected packet, the Camelyon17 certificate explicitly records `selection_uses_external_metrics=false`.

5 Benchmarks and Evidence

The current artifact package uses two official frozen-image benchmark families and one public locked-split sensor benchmark. Waterbirds is a spurious-correlation benchmark in which bird type is the target and background/place is the source. Camelyon17 is a medical hospital-shift benchmark in WILDS [5]; it is used only as a representation-reliability benchmark, not as clinical evidence. GaitPDB is the PhysioNet Gait in Parkinson’s Disease dataset; we use deterministic time-series features, patient/control as the target, and

Method	Target BA	Worst group	Source leakage
Waterbirds VERA	0.8012 ± 0.0005	0.6234 ± 0.0018	0.8935 ± 0.0005
Waterbirds reweighted ERM	0.8685 ± 0.0007	0.8186 ± 0.0012	0.8935 ± 0.0004
Camelyon17 VERA	0.8905 ± 0.0000	0.8464 ± 0.0000	val. 0.8180
Camelyon17 GroupDRO probe	0.8790 ± 0.0000	0.8176 ± 0.0000	val. 0.8180
GaitPDB VERA	0.6944 ± 0.0000	0.6667 ± 0.0000	val. 0.7606
GaitPDB GroupDRO probe	0.6650 ± 0.0000	0.5556 ± 0.0000	val. 0.7606

Table 2: Artifact-backed benchmark summary. BA denotes balanced accuracy. On Waterbirds, VERA abstains/no-edits because the candidate edit frontier does not certify a better safe edit; this is a valid VERA decision, not an accuracy win claim. On Camelyon17, VERA preserves stronger external target and worst-group metrics than the GroupDRO-style probe under the frozen representation protocol. GaitPDB is a public locked-split sensor benchmark and is not counted as an official challenge row.

Waterbirds MANCE++ metric	Before	After	Delta
External source leakage BA	0.9059	0.4468	-0.4591 ± 0.0067
External target BA	0.7390	0.6915	-0.0475 ± 0.0025
External worst target-source acc.	0.4907	0.5283	0.0377 ± 0.0054
Validation source leakage BA	0.9065	0.4523	-0.4543 ± 0.0116
Validation target BA	0.7457	0.6709	-0.0748 ± 0.0079

Table 3: Five-seed official-code MANCE++ reference run on full Waterbirds splits. Values are means; delta intervals are 95 percent confidence intervals over seeds.

acquisition-site code as the source. The gait row is public and subject-split, but not an official challenge split. For Camelyon17, we additionally run a full 455,954-example source-direction projection frontier. The certificate is selected using validation metrics only: with target-loss budget 0.02 and source-reduction target 0.05, no projection strength certifies source reduction, so VERA returns **ABSTAIN**. External target and worst-group metrics are reported after the decision; external source leakage is scoped out because the binary source encoding has a single source class in the external split.

6 Reference MANCE++ Baseline

Because MANCE++ is a recent and close nonlinear erasure baseline, we integrated the official upstream implementation into the VERA harness rather than relying only on a proxy. The Waterbirds run is full-split and five-seed. MANCE++ reduces external source leakage balanced accuracy from 0.9059 to 0.4468 with mean delta -0.4591 ± 0.0067 . External target balanced accuracy changes from 0.7390 to 0.6915, and worst target-source accuracy improves from 0.4907 to 0.5283. These numbers support the claim that MANCE++ is a strong erasure candidate, but they also show why VERA’s decision layer is necessary: leakage reduction and target preservation must be adjudicated under explicit constraints.

For Camelyon17, the official MANCE++ reference implementation was run on the full no-cap frozen-representation store: 302,436 training, 68,464 validation, and 85,054 external examples. Validation source leakage balanced accuracy decreased from 0.8492 to 0.5635, while validation target balanced accuracy decreased from 0.8974 to 0.8876. External target balanced accuracy decreased from 0.8905 to 0.8741, and external worst target-source accuracy decreased from 0.8465 to 0.8137. This is a claim-grade official-code Camelyon17 MANCE++ reference row under the frozen-representation protocol. External source leakage remains blank because the binary source encoding has a single source class in the external split.

7 Reference Linear Eraser Rows

Because INLP and LEACE are cheap on frozen features, they are no longer left as proxy-only comparisons. The local packet includes real INLP rows and official LEACE rows using the pinned `concept-erasure`

Baseline	Upstream status	Current claim status
MANCE++	Official code pinned and run	Waterbirds and Camelyon references
INLP	Local closed-form row run	Real frozen-feature reference row
LEACE	Official code pinned and run	Official frozen-feature reference row
R-LACE	Official code pinned	Proxy until matched receipt
TaCo	Official code pinned	Proxy until matched receipt
SPLINCE/SPLICE	No pinned official repo	Proxy or paper-only boundary

Table 4: Reference-baseline boundary recorded by the upstream inventory audit. Pinned code means the official repository and commit are materialized locally; it does not by itself create reference parity without a matched VERA receipt. INLP and LEACE now have matched frozen-feature receipts in `linear_eraser_reference_report.json`; R-LACE, TaCo, and SPLINCE/SPLICE remain scoped future reference-parity work.

Dataset	Method	Target BA	Worst group	Leak val.
Waterbirds	No edit	0.7560	0.3988	0.9065
Waterbirds	INLP rank 4	0.7778	0.4455	0.8365
Waterbirds	LEACE official	0.5954	0.0701	0.4821
Camelyon17	No edit	0.8905	0.8464	0.8492
Camelyon17	INLP rank 4	0.8943	0.8628	0.8259
Camelyon17	LEACE official	0.8891	0.8427	0.5015
GaitPDB	No edit	0.6944	0.6667	0.7606
GaitPDB	INLP rank 4	0.7042	0.6667	0.6152
GaitPDB	LEACE official	0.7239	0.6667	0.5515

Table 5: Real linear eraser rows from `research/artifacts/linear_eraser_reference_report.json`. Lower source leakage is better, but target and worst-group damage determine whether an eraser is deployable.

implementation. These rows are not used to claim universal erasure state of the art; they are used to show the target/source tradeoff that VERA is designed to audit.

The linear rows strengthen the submission because they address the cheapest close baselines directly. LEACE sharply reduces Waterbirds leakage but collapses worst-group accuracy, so an unconditional “erase more” objective is unsafe. On Camelyon17, INLP rank 4 slightly improves target and worst-group metrics while reducing validation leakage, whereas LEACE reduces leakage more aggressively with a small target/worst-group cost. VERA’s role is to expose that tradeoff under a registered contract and abstain when the contract is not certified.

8 Abstention

VERA’s differentiating behavior is the ability to refuse an edit. The synthetic abstention certificate demonstrates the geometry of overlapping target and source directions, where every leakage-reducing candidate violates the target constraint. The real benchmark stress tests apply the same logic to artifact backed results. This matters for main-track review because the protocol is not merely selecting the best-looking point in a table; it records when the table does not justify deployment.

The full Camelyon17 projection frontier gives a concrete medical benchmark example. Across strengths 0 through 1, the validation target-loss upper confidence bound remains below 0.02, but the source-reduction lower confidence bound is 0 for every candidate. VERA therefore abstains: the candidate family appears target-safe, but it fails the pre-registered source reduction requirement. This is exactly the behavior the protocol is designed to expose.

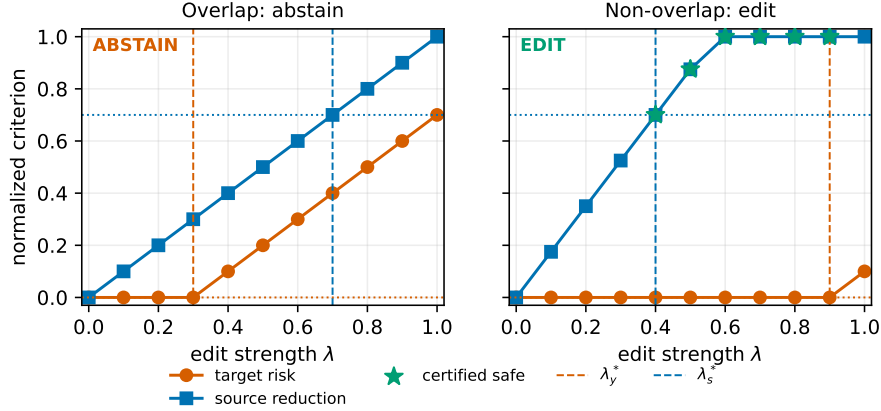


Figure 2: Synthetic abstention geometry. VERA abstains when the candidate frontier cannot certify both target preservation and source leakage reduction.

9 Reproducibility and Claim Boundaries

Every reported benchmark row is backed by JSON receipts, per-seed CSV files, statistical reports, exact paired sign-test addenda, and audit scripts. With five seeds, the exact two-sided sign tests are best interpreted as directionality checks because their smallest possible nonzero p-value is 0.0625. The main readiness gate can therefore verify integrity, but it should not be read as a conventional large-sample significance claim.

The most important limitation is that VERA is a selection and abstention protocol, not a universal erasure method. It depends on candidate erasers, source labels, target labels, and a meaningful validation split. External shift remains a diagnostic readout unless additional bounded-shift assumptions are justified. Full-reference MANCE++ evidence is claim-grade for Waterbirds and Camelyon17, and the packet now includes real INLP plus official LEACE frozen-feature rows. The manuscript should therefore avoid any statement that VERA is universally state of the art at nonlinear erasure or clinically deployable in medical settings.

The strongest reviewer objection is that VERA could be mistaken for an erasure algorithm and then judged as if it must dominate every modern eraser. The paper does not make that claim. VERA’s contribution is the validation-only edit-or-abstain decision rule, the receipt format, and the claim boundary around candidate edits. Reference implementations are used when matched receipts exist; otherwise, the manuscript labels rows as proxy stress tests and denies universal erasure state-of-the-art claims.

10 Code Availability

The release repository is <https://github.com/RudraChopra/vera-edit-or-abstain>. It contains the paper sources, audit scripts, configuration manifests, small receipts, and reproduction commands needed to inspect the reported VERA claims. Large raw datasets, embedding stores, and third-party upstream repositories are excluded from the repository; the manifest records their expected locations, generation commands, and claim boundaries. Anonymous conference submissions should use the de-identified supplemental archive rather than the named repository link during review.

11 Conclusion

VERA reframes representation editing as a decision problem: construct a frontier, certify target preservation and leakage reduction, then edit or abstain. This framing is compatible with linear erasers, nonlinear manifold erasers, and domain-shift benchmarks. The current evidence package is strong enough to support a method-paper draft with explicit boundaries, while the next scientific step is to deepen the theory and broaden full-reference erasure baselines beyond MANCE++, INLP, and LEACE.

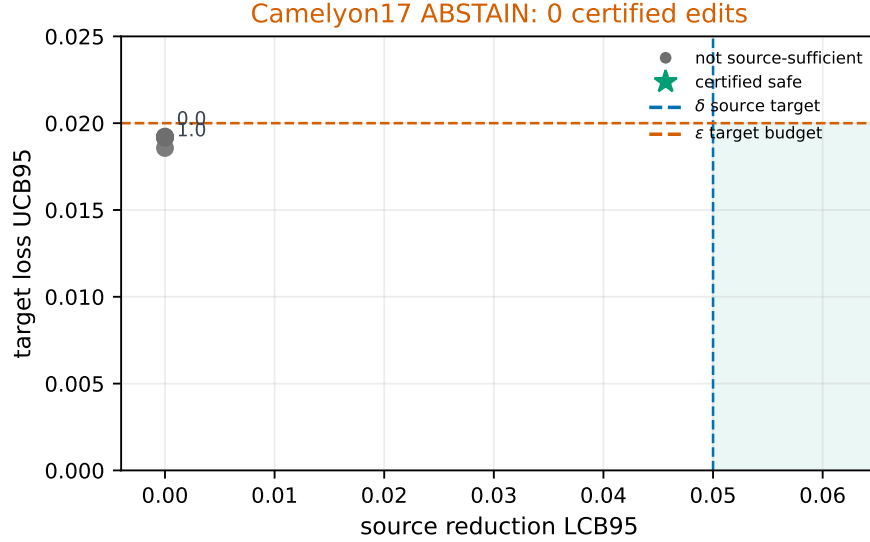


Figure 3: Full Camelyon17 projection frontier certificate. The green feasible region requires both source-reduction and target-preservation bounds. No candidate enters the feasible region, so VERA returns ABSTAIN.

References

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